

SplitFed Speech Emotion Classification

Multiseed Experiment Tables

Seeds 42–46

All values are the arithmetic mean $\pm$ sample standard deviation over five independent runs. Each seed controls model initialization, actor-disjoint data splits, client and server runtimes, and MergeSFL cohort search.

Table 1: E1 cross-corpus anger F1 after 50 local epochs. Each model is trained only on the corpus named in its row.

Train model	CREMA-D test	RAVDESS test	SAVEE test
CREMA-D	0.6860 ± 0.0337	0.1231 ± 0.1337	0.3304 ± 0.2247
RAVDESS	0.4370 ± 0.0946	0.6601 ± 0.1116	0.2326 ± 0.1837
SAVEE	0.2650 ± 0.0909	0.1584 ± 0.1111	0.2985 ± 0.2731

Table 2: E2 centralized versus FedAvg quality. Within each run, metrics are sample-weighted across the three corpus-specific test sets.

Method	Accuracy	Anger F1	Macro F1	Precision	Recall	PR-AUC
Centralized	0.8917 ± 0.0128	0.6625 ± 0.0284	0.7983 ± 0.0170	0.7110 ± 0.0517	0.6594 ± 0.0760	0.7805 ± 0.0177
FedAvg	0.8939 ± 0.0214	0.6475 ± 0.0401	0.7907 ± 0.0288	0.7453 ± 0.0250	0.6308 ± 0.0339	0.7525 ± 0.0221

Table 3: E3 protocol quality at round 50. Within each run, metrics are sample-weighted across corpus-specific client evaluations.

Method	Accuracy	Anger F1	Macro F1	Precision	Recall	PR-AUC
MergeSFL	0.8518 ± 0.0454	0.6021 ± 0.0232	0.7526 ± 0.0288	0.6086 ± 0.0928	0.6884 ± 0.1087	0.7221 ± 0.0423
SFLv1	0.8648 ± 0.0425	0.6518 ± 0.0355	0.7799 ± 0.0364	0.6222 ± 0.0601	0.7515 ± 0.0704	0.7601 ± 0.0101
SFLour	0.8710 ± 0.0390	0.6459 ± 0.0240	0.7812 ± 0.0265	0.6514 ± 0.0819	0.6989 ± 0.0875	0.7544 ± 0.0174

Table 4: E4 SFLour workload-policy quality at round 50. Metrics are sample-weighted across corpus-specific client evaluations within each run.

Policy	Accuracy	Anger F1	Macro F1	Precision	Recall	PR-AUC
Full local epoch	0.8710 ± 0.0390	0.6459 ± 0.0240	0.7812 ± 0.0265	0.6514 ± 0.0819	0.6989 ± 0.0875	0.7544 ± 0.0174
Fixed steps	0.8759 ± 0.0240	0.6401 ± 0.0205	0.7809 ± 0.0194	0.6595 ± 0.0659	0.6548 ± 0.0328	0.7201 ± 0.0336
Balanced	0.8501 ± 0.0404	0.6202 ± 0.0235	0.7599 ± 0.0260	0.5880 ± 0.1001	0.7105 ± 0.1159	0.7111 ± 0.0080

Table 5: E4 workload-policy cost. CPU covers training roles; transport uses binary GiB. Batches/round are CREMA-D/RAVDESS/SAVEE.

Policy	Batches/round	Wall (s)	CPU user (s)	CPU sys (s)	Transport (GiB)	Messages
Full local epoch	737/143/45	805.77 ± 43.15	1700.48 ± 101.81	313.02 ± 17.12	22.8629 ± 0.0094	95114 ± 49
Fixed steps	143/143/143	300.22 ± 19.69	859.32 ± 55.02	127.79 ± 7.76	10.7720 ± 0.0005	45554 ± 5
Balanced	64/64/45	181.99 ± 11.84	359.62 ± 23.16	57.60 ± 3.53	4.5312 ± 0.0005	19954 ± 5

Table 6: E1–E4 runtime and communication cost under Resource Metrics v2. E1 has no end-to-end wall interval. CPU times are summed across training roles; transport uses binary GiB.

Method	Wall (s)	CPU user (s)	CPU sys (s)	Transport (GiB)	Messages
E1 Local	–	352.55 ± 18.57	81.25 ± 1.42	0.0000 ± 0.0000	0 ± 0
E2 Centralized	486.81 ± 30.54	362.02 ± 23.72	82.34 ± 1.67	0.0000 ± 0.0000	0 ± 0
E2 FedAvg	562.80 ± 39.38	2798.90 ± 118.79	154.48 ± 7.41	1.9590 ± 0.0000	300 ± 0
E3 MergeSFL	269.94 ± 33.63	348.81 ± 55.06	91.30 ± 8.05	7.3202 ± 0.0005	31524 ± 5
E3 SFLv1	896.97 ± 52.32	4289.21 ± 212.27	369.07 ± 22.36	22.8629 ± 0.0094	95264 ± 49
E3 SFLour	805.77 ± 43.15	1700.48 ± 101.81	313.02 ± 17.12	22.8629 ± 0.0094	95114 ± 49
E4 Balanced	181.99 ± 11.84	359.62 ± 23.16	57.60 ± 3.53	4.5312 ± 0.0005	19954 ± 5
E4 Fixed steps	300.22 ± 19.69	859.32 ± 55.02	127.79 ± 7.76	10.7720 ± 0.0005	45554 ± 5

Table 7: Topology-aware owned tensor footprint in binary MiB. Max client is the requirement for one client node; sums are logical footprints over isolated participants, not process RSS.

Method	Client max	Clients sum	Central trainer	Split max	Split sum	FedServer	Logical total
E1 Local	125.55 ± 0.06	196.34 ± 0.06	–	–	–	–	196.34 ± 0.06
E2 Centralized	–	–	192.89 ± 0.00	–	–	–	192.89 ± 0.00
E2 FedAvg	153.40 ± 0.00	232.93 ± 0.00	–	–	–	6.68 ± 0.00	239.62 ± 0.00
E3 MergeSFL	133.51 ± 0.00	173.26 ± 0.00	–	6.55 ± 0.00	6.55 ± 0.00	0.13 ± 0.00	179.94 ± 0.00
E3 SFLv1	133.77 ± 0.00	174.05 ± 0.00	–	19.63 ± 0.00	58.88 ± 0.00	0.13 ± 0.00	233.07 ± 0.00
E3 SFLour	133.77 ± 0.00	174.05 ± 0.00	–	19.63 ± 0.00	19.63 ± 0.00	0.13 ± 0.00	193.81 ± 0.00
E4 Balanced	133.77 ± 0.00	174.05 ± 0.00	–	19.63 ± 0.00	19.63 ± 0.00	0.13 ± 0.00	193.81 ± 0.00
E4 Fixed steps	133.77 ± 0.00	174.05 ± 0.00	–	19.63 ± 0.00	19.63 ± 0.00	0.13 ± 0.00	193.81 ± 0.00

Table 8: Maximum per-process RSS/CUDA allocated memory by runtime role in binary MiB. Each cell is (RSS mean $\pm$ SD)/(CUDA mean $\pm$ SD). Peaks from different processes are not summed.

Method	Client	Central trainer	SplitServer	FedServer
E1 Local	$(2024.76 \pm 8.96)/(149.35 \pm 0.00)$	–	–	–
E2 Centralized	–	$(2076.78 \pm 11.20)/(149.35 \pm 0.00)$	–	–
E2 FedAvg	$(2052.89 \pm 9.53)/(149.35 \pm 0.00)$	–	–	$(805.03 \pm 0.73)/(2.25 \pm 0.00)$
E3 MergeSFL	$(1839.30 \pm 20.67)/(10.73 \pm 0.00)$	–	$(1553.50 \pm 3.87)/(177.22 \pm 0.00)$	$(766.67 \pm 0.89)/(0.14 \pm 0.00)$
E3 SFLv1	$(1853.72 \pm 14.52)/(11.00 \pm 0.00)$	–	$(1624.74 \pm 50.44)/(142.42 \pm 0.00)$	$(736.08 \pm 41.46)/(0.14 \pm 0.00)$
E3 SFLour	$(1843.81 \pm 11.62)/(11.00 \pm 0.00)$	–	$(1629.74 \pm 3.30)/(239.07 \pm 0.00)$	$(766.23 \pm 0.94)/(0.14 \pm 0.00)$
E4 Balanced	$(1843.16 \pm 19.31)/(11.00 \pm 0.00)$	–	$(1623.17 \pm 2.56)/(239.07 \pm 0.00)$	$(766.13 \pm 1.43)/(0.14 \pm 0.00)$
E4 Fixed steps	$(1837.43 \pm 14.59)/(11.00 \pm 0.00)$	–	$(1623.54 \pm 3.70)/(239.07 \pm 0.00)$	$(765.32 \pm 1.32)/(0.14 \pm 0.00)$

Table 9: Per-client E1 local-training resources. Memory uses binary MiB; evaluation runs in separate processes and is excluded.

Corpus	Epoch time sum (s)	Owned	Dataset	Process RSS	Phase RSS	CUDA alloc.	CUDA reserved
CREMA-D	331.32 ± 12.54	125.55 ± 0.06	105.52 ± 0.06	2024.76 ± 8.96	2025.97 ± 8.89	147.32 ± 0.00	216.00 ± 0.00
RAVDESS	65.81 ± 2.63	41.56 ± 0.00	21.54 ± 0.00	1904.65 ± 4.69	1905.92 ± 4.59	147.58 ± 0.00	216.00 ± 0.00
SAVEE	21.44 ± 0.79	29.24 ± 0.00	9.21 ± 0.00	1895.31 ± 3.11	1896.41 ± 3.02	149.35 ± 0.00	218.00 ± 0.00